

An **integrated insurance framework** designed to quantify **AI-driven cascading risks** and cover financial losses in **pharma smart factories**.

Target Innovative Pharma & Biotech firms, AI-driven drug discovery startups, and CDMOs transitioning to cloud-based Smart Manufacturing

Problem Definition

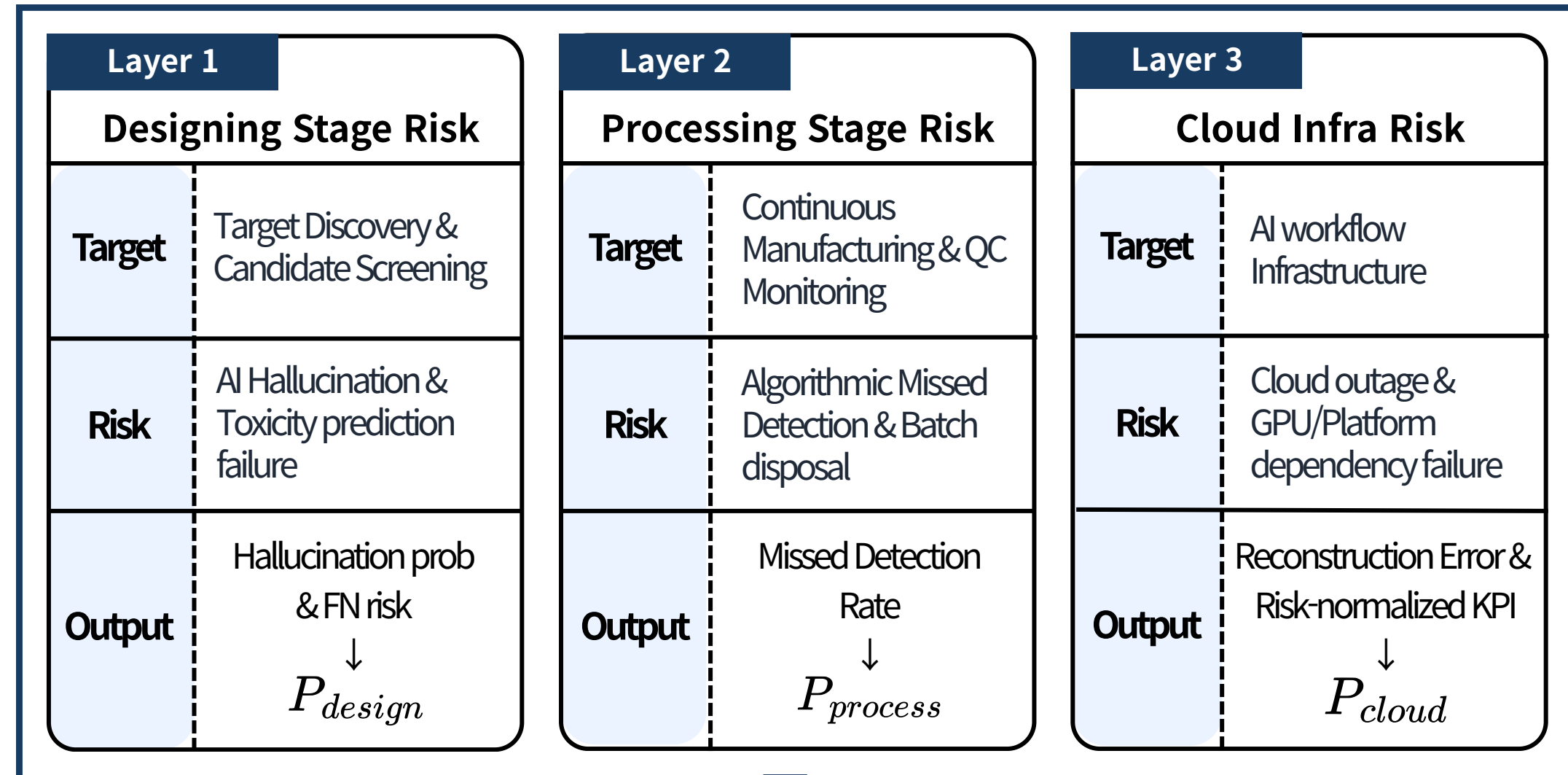
- Pharma risks are shifting from **Physical Accidents** to **Data & Algorithm-driven Failures**
- AI drug discovery market is booming (CAGR 25~30%, future value \$100B), yet fundamentally exposed to **3 Core Uncertainties**: 1) **AI Hallucination**, 2) **AI Detection Failure**, and 3) **Cloud Cascading Failures**
- Impact**: Massive financial losses, compromised public safety, and degraded competitiveness

**Note: (CAGR) Compound Annual Growth Rate

Current Situation Analysis

Category	As-Is (Current)	To-Be (AEGIS-Rx)
Cyber Insurance	External attacks (Hacking, Ransomware)	Internal system failures (AI Hallucination & Logic Errors)
Business Interruption Insurance	Shutdowns via physical damage (Fire, Earthquake)	Software-driven downtime & Batch disposal
Cloud SLA	Limited service credit compensation	Direct operational & opportunity loss coverage

→ Existing insurance frameworks **do not sufficiently cover** AI-driven pharmaceutical operational risks



Layer 4 Integrated Loss & AI Risk Score Calculation

EQUATIONS

$$ExpectedLoss = P_{design} \times L_{R\&D} + P_{process} \times L_{batch} + P_{cloud} \times L_{operation} + (P_{process} \times P_{cloud}) \times L_{batch}$$

$$RiskScore = \alpha P_{design} + \beta P_{process} + \gamma P_{cloud} + \delta (P_{process} \times P_{cloud})$$

Layer 5 AEGIS-Rx (AI-Embedded Guarantee Insurance System for Rx)

Expected Loss → **Insurance Premium**, RiskScore → **Risk Underwriting**

AI Solution

Predictive parametric insurance detects and scores loss-related risks in real time, linking them to automated triggers and shifting insurance from ex-post compensation **to proactive risk response**.

TRIGGER and PAYOUT

Coverage 1
(Indemnity)

- **Trigger:** AI candidate defect halts project
- **Payout:** R&D sunk costs

Coverage 2
(Indemnity)

- **trigger:** AI misses process anomaly → defective batch
- **Payout:** disposal & recall costs

Coverage 3
(Parametric)

- **trigger:** cloud delay/load index exceeds threshold
- **Payout:** immediate indexed compensation

EXECUTION PROCESS

STEP1: Underwriting & Audit

Inspect AI model architecture and training data defects → **Deductible, Coverage limit**

STEP2: Dynamic Pricing

Periodic AI audit reports

→ Relaxation of exclusions and expansion of coverage based on management levels

STEP3: Claims Adjustment

Identify the true cause of the accident (human error vs. AI)

PAYOUT SCENARIO

Scenario	Layer
Errors found during the design phase	L1
Design errors transfer to production	L1+L2
Batch disposed due to AI failure(FN)	L2
Cloud trigger triggered, no batch disposed	L3
Cloud trigger triggered + batch discarded	L2+L3

Executive Strategy

PILOT

[Risk] FN rate based
[Industry] Pharmaceutical Industry

SCALE-UP

[Risk] Comprehensive evaluation criteria
[Industry] Logistics Stage and entire smart factory industry

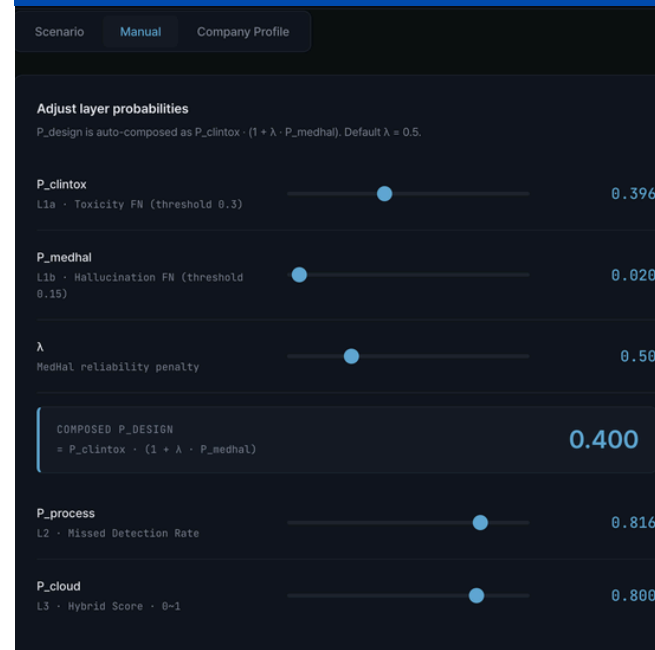
EXPECTED EFFECTS

- Managing risks using Integrated Indemnity + Index Insurance (**basis risk ↓**)
- Pharmaceutical Company: hedging recall/litigation costs → **minimizing financial volatility**
- Insurance Company: Securing a leading position in the **non-existent B2B deep tech insurance market** → creating new revenue streams

OBSTACLES & MITIGATION

- Predicted loss ≠ Actual payout → Set **policy limits and deductibles based on MFL**
- Ambiguous restoration criteria → Define recovery completion, business interruption period, and return-to-normal operation in advance
- Shared cloud/AI dependency risk → Manage **portfolio-level concentration** & secure financial stability through **reinsurance**
- **Data governance** as execution backbone → Pre-structure data, recovery, and compensation standards

Risk Score SIMULATOR



SOCIAL VALUE

- Risk-sharing for AI prediction failures & **End-to-end AI protection** → Faster AI adoption and drug development/ Digital transformation across R&D, manufacturing, and quality control
- Expanded pharma role → From drug manufacturer to **data- and algorithm-driven treatment solution provider**
- Broader industry impact → Stronger **supply chain stability**, faster smart factory adoption, and higher brand trust
- Insurance also serves **proactive risk management** → helping establish guidelines for AI safety and reliability
- Considerations for Sustainability: Blame-shifting, Concept Drift

Top Risk Rankings in the Chemical, Pharmaceutical, and Biopharma Sectors

Rank	2016	2018	2020	2022
#1	Business interruption & supply chain disruption	Business interruption & supply chain disruption	Business interruption & supply chain disruption	Business interruption & supply chain disruption
#2	Market stagnation & shrinkage	Market stagnation & shrinkage	Cyber crime & IT failure	Cyber crime & IT failure
#3	Loss of reputation & brand value	Natural disasters	Changes in legislation & regulation	Energy crisis
#4	Macroeconomic changes	Market stagnation & shrinkage	Product recall, etc.	Changes in legislation & regulation
#5	Changes in legislation & regulation	Changes in legislation & regulation	Fire & explosion	Macroeconomic changes

**Note: The 2018 data represents all industries
**Source: Allianz Risk Barometer Top Business Risk (2016~2023)

While **business interruption** has consistently remained the dominant risk in the pharmaceutical sector, **cyber incidents and IT failures** have rapidly emerged as critical new threats. This shift highlights the growing need for comprehensive insurance frameworks capable of quantifying AI, operational monitoring risks, and cloud.

Industry Adoption Cases

- (1) **Sanofi** | Declared an “**All in on AI**” strategy to revolutionize the entire R&D and candidate screening lifecycle
- (2) **Pfizer** | Actively expanding external **AI collaborations** for target discovery and molecule optimization
- (3) **AstraZeneca** | Deployed **Digital Twins** for manufacturing planning and continuous process simulation
- (4) **SK Bioscience** | Implemented the **AI-based ‘ADO System’** for advanced manufacturing process optimization

Insurance Gap

- Traditional cyber insurance focuses on **external cyberattacks** (ransomware, hacking, external data breaches)
 - However, AI-enabled pharmaceutical operations introduce **cascading risks** that defy existing frameworks, characterized by:
 - Limited historical loss data** for pricing
 - Complex accountability attribution** for AI-assisted decisions
 - Massive, multi-company vulnerabilities** caused by shared cloud/GPU infrastructure
- Significant **protection gaps** remain, leaving AI-integrated pharma operations critically exposed to **unhedged financial losses**

Regulatory Trend

FDA (USA)

- Recognizes **AI-assisted drug development workflows**
- Requires **traceable source data, bias mitigation, lifecycle human oversight, explainability and validation**

EMA (EU)

- Adopts a **risk-based validation approach**
- Higher validation requirements for AI systems affecting **clinical outcomes**

MFDS (KOR)

- No formal AI drug development guideline** yet
- However, the MFDS has demonstrated **growing interest in AI adoption**

→ Global agencies are increasingly moving toward **AI-integrated pharmaceutical ecosystems** rather than restricting AI adoption itself

04 Appendix B — Layer 1: Design Risk Evidence

MedHal

Objective

To quantify the risk that generative AI produces hallucinated medical answers or fails to detect incorrect information, and convert it into an AI decision risk score.

Dataset

Medhallu dataset(Medical Halluciation Answering dataset)

- The dataset includes medical questions, reference knowledge, ground-truth, hallucinated answers, difficulty levels, and hallucination categories.
- It is designed to **evaluate the ability of large language models to detect hallucinations** in medical question-answering tasks.

Modeling

Sentence-Bert + XGBoost Classifier

- Step1:** Convert Q/K/GT/Answer into **sentence embeddings**
- Step2:** Generate **similarity features** and risk-related variables
- Step3:** Train classifier to estimate hallucination probability and FN ratio.

Outputs

At threshold 0.15,

- FN Risk(P_{Medhal}) = 0.02
- Sensitivity (Recall) = 0.98

In the design phase, **since recall for detecting FN is more important** than accuracy, the **threshold** for each data point was selected **based on the F2 score**.

$$F2Score = 5 \times \frac{Precision + Recall}{(4 \times Precision) + Recall}$$

ClinTox

Objective

To quantify the False Negative (FN) rate, where the AI model incorrectly classifies toxic compounds as non-toxic.

Dataset

ClinTox Dataset (Clinical Trial Toxicity)

- The dataset contains information on approximately 1,478 compounds, including labels for FDA approval status and actual clinical trial toxicity outcomes (CT_TOX)
- Data Imbalance: Toxic compounds account for only about 6.8% of the dataset
→ **Design strategy:** Instead of accuracy, '**AUC**' and '**Sensitivity**' were selected as the key evaluation metrics

Modeling

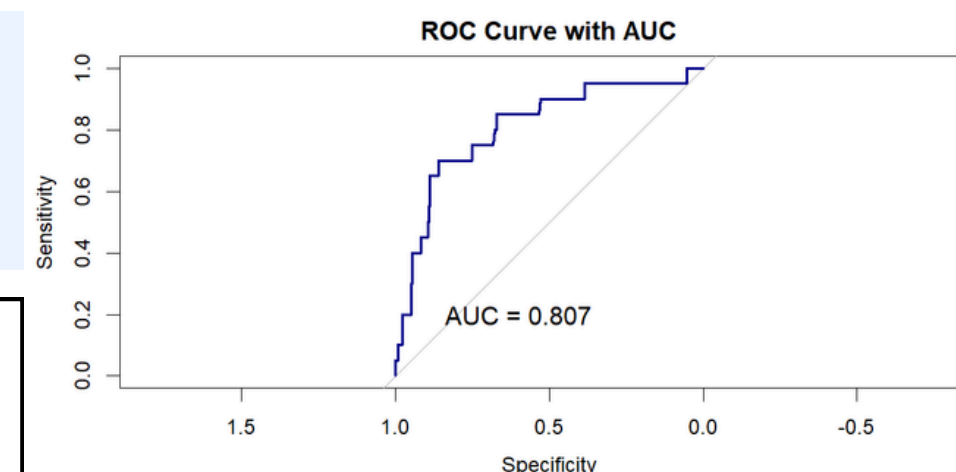
XGBoost Classifier

An ensemble model optimized for handling high-dimensional molecular data.

- Step1:** The molecular structure (SMILES) is transformed into a 1024-bit binary Morgan Fingerprint vector
- Step2:** The classification threshold was lowered from the default value of 0.5 to 0.3

Outputs

- AUC = 0.807
- FN Risk(P_{ClinTox}) = 0.80
- Sensitivity (Recall) = 0.20



$$P_{\text{design}} = P_{\text{ClinTox}} \times (1 + \lambda P_{\text{Medhal}})$$

Defined as a toxicity-based design risk **adjusted by AI decision reliability**.
The parameter λ is set as a **sensitivity coefficient**, with 0.5 used as the baseline scenario.

Appendix C — Layer 2 + Layer 3: Process & Cloud Risk Evidence

Chemical Process

Objective

To quantify the probability of the AI failing to detect subtle faults in pharmaceutical manufacturing

Dataset

Chemical Process Monitoring Time Series Dataset

- **Synthetic multivariate** industrial sensor dataset (**1-minute interval** sampling), simulating gradual fault progression, sensor noise, and operational variability
- Only **Regime A (baseline operating condition)** was used for pilot anomaly detection modeling

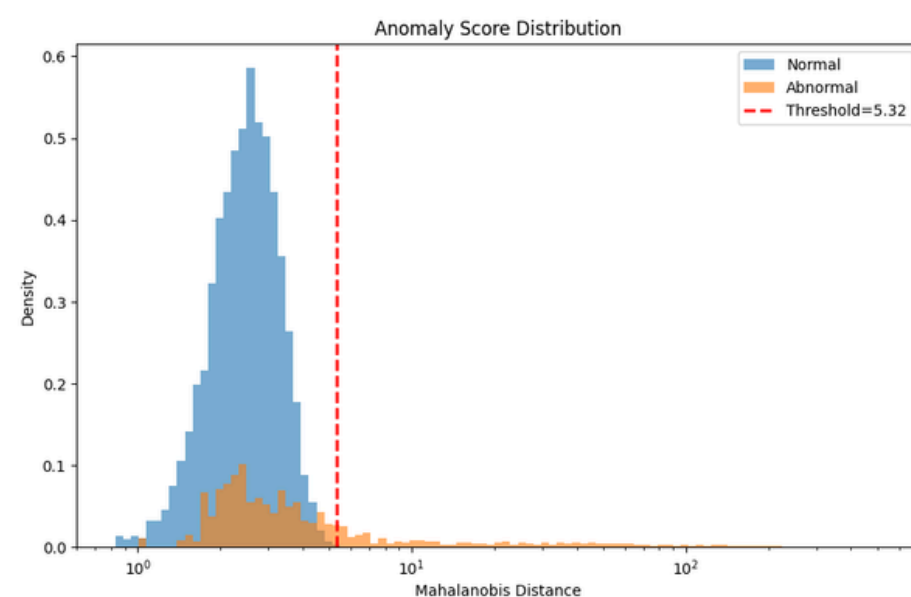
Modeling

LSTM-Autoencoder + Mahalanobis Distance

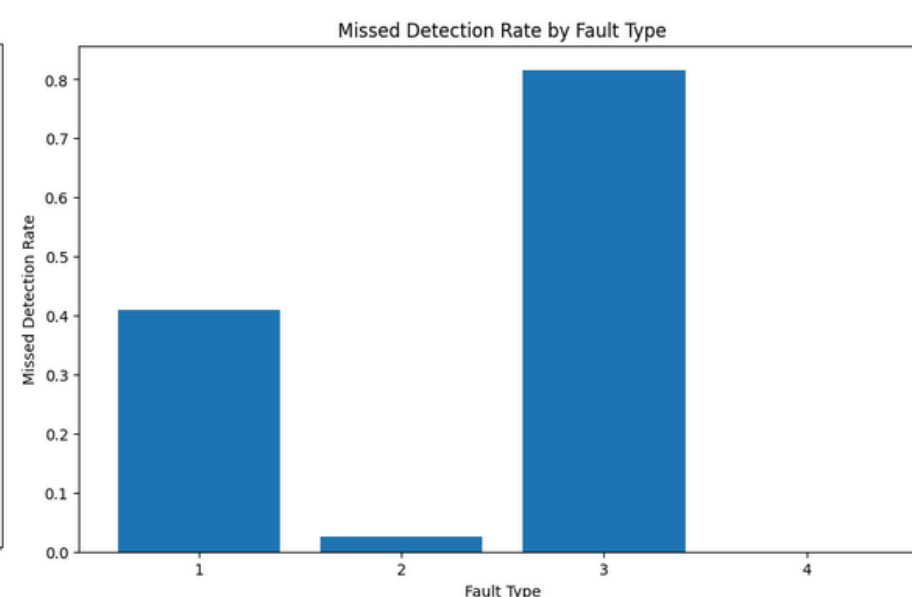
- **Step 1:** Captures **time-series dynamics** to learn and reconstruct normal process behaviors
- **Step 2:** Calculates the final **Anomaly Score** by incorporating **covariance structures** among multivariate sensors, advancing beyond simple reconstruction error

Outputs

P_{process} (Monitoring failure probability)



→ Selected a **99th-percentile threshold** to suppress false alarms (<1%) for operational continuity, resulting in a **16.01% overall missed detection rate**



→ **Sensor drift faults (Type 3)** showed the highest missed detection rate (**81.61%**), providing a basis for future **fault-specific underwriting** and **premium differentiation strategies**

Cloud Infra Risk

Objective

Quantify cloud infrastructure risk that disables AI monitoring;
Convert to parametric insurance trigger.

Dataset

AIOps Challenge 2020 (Tsinghua/CCF)

1. Real-world cloud microservice traces with labeled fault events (1min interval);
2. extracted 5 KPIs aligned with incident labels for trigger derivation

Modeling

Hybrid Score: Rule-based + LSTM Autoencoder

- **Step 1:** Rule-based scoring (60%)
- weighted sum of 5 KPIs for explainable baseline
- **Step 2:** LSTM Autoencoder (40%)
- learns normal patterns, flags reconstruction errors

$$P_{\text{cloud}} = 0.6 \times \text{Rule} + 0.4 \times \text{LSTM}$$

(weighted avg, not max, to suppress false alarms)

Outputs

P_{cloud} (Cloud Risk Score)

Trigger: $P_{\text{cloud}} \geq 0.55$
for ≥ 2 min

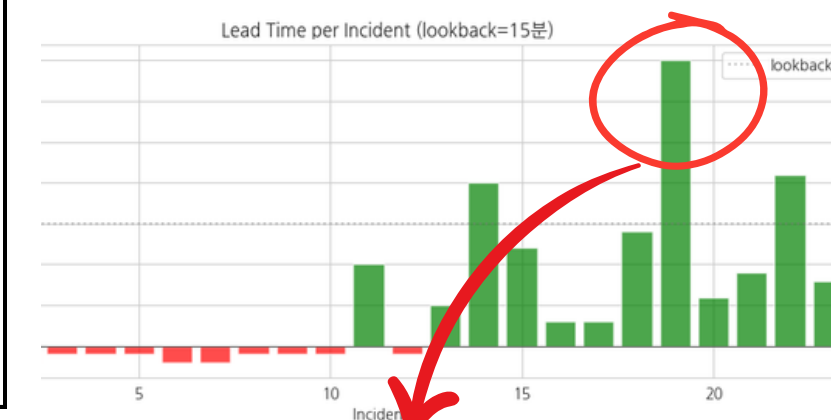
Performance : Event Recall: 65.7%

Avg Lead Time: +5.5 min

ROC AUC: 0.78

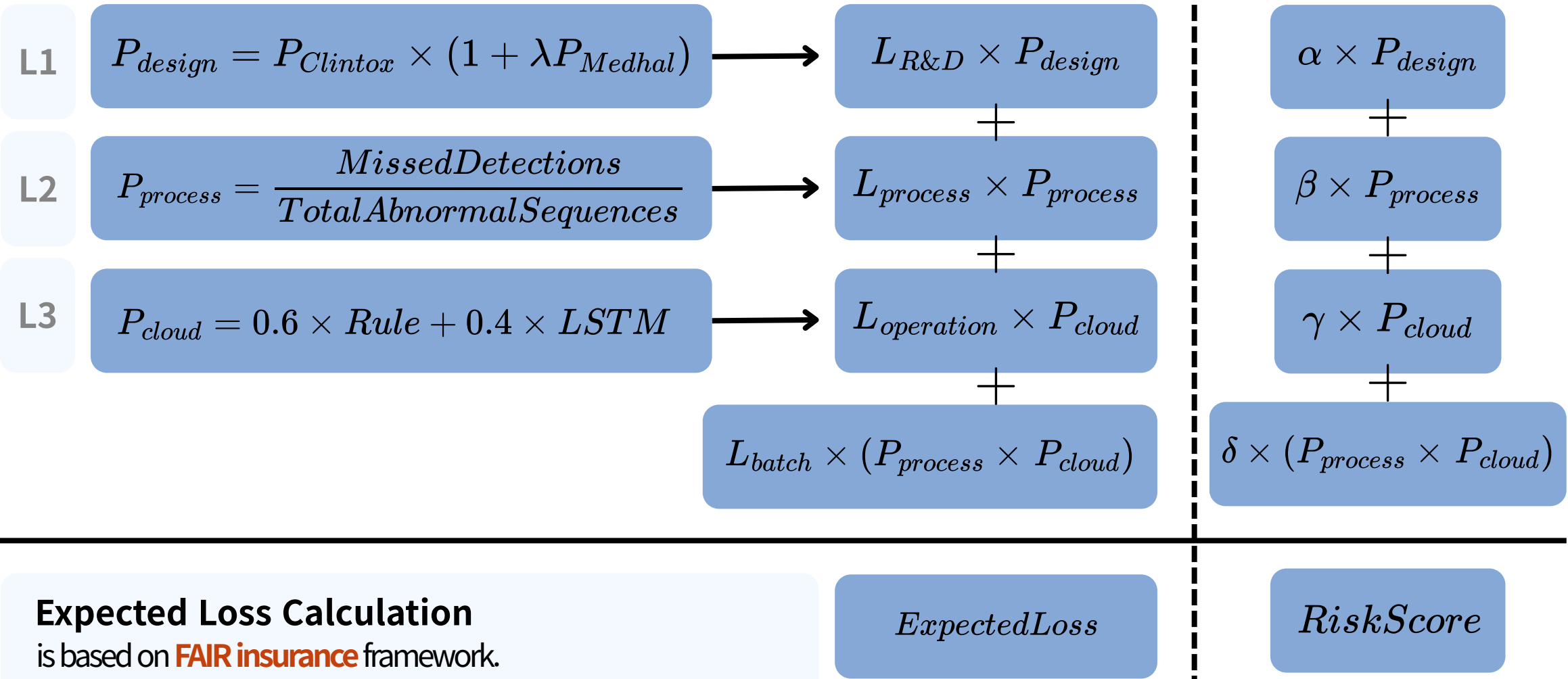
Difference

- Captures AI's operational environment, not AI errors themselves (unlike L1/L2)
 - Hybrid model balances explainability (Rule) + detection (LSTM), beyond single-model approaches
 - Trigger: IF $P_{\text{cloud}} \geq \theta$ for $\geq N$ min THEN payout - θ , N from data-driven grid search
 - **Lead Time:** trigger fires + N min before incident
- ex-post compensation + ex-ante warning



Trigger fires +5.5 min before incidents on average (12/23 with positive lead time)

Appendix D — Layer 4 + 5: Risk Calculation & Execution Process



Risk score coefficient criteria : based on ICH Q9 Criteria (International Council for Harmonisation for Quality Risk Management)

$\alpha = 0.2Q + 0.3S + 0.3R + 0.0D + 0.2F + 0.0O$

$\beta = 0.2Q + 0.2S + 0.2R + 0.0D + 0.4F + 0.0O$

$\gamma = 0.0Q + 0.0S + 0.0R + 0.6D + 0.0F + 0.4O$

$\delta = 0.1Q + 0.1S + 0.1R + 0.3D + 0.2F + 0.2O$

*The coefficient was calculated arbitrarily and is subject to change depending on the insurer's risk assessment criteria.

	Impact	Major Risk
Q	Quality	Design, Process
S	Safety	Design
R	Regulatory/GMP	Design
D	Data Integrity	Cloud
F	Financial	Process
O	Operational	Cloud

EXECUTION PROCESS

STEP1: Underwriting & Audit

AI model, training data, FN rate, cloud architecture audit
→ **Initial Risk Score + Expected Loss**
→ Deductible & Coverage Limit

EXAMPLE

Risk	Grade	Insurance Underwriting
0-20	A	Standard Acceptance
20-40	B	Slight surcharge
40-60	C	Conditional Acceptance
60-80	D	High Deductible/Partial Coverage
80-100	E	Acceptance Restriction/High-risk contracts

STEP2: Dynamic Pricing

Periodic AI audit report + cloud/process monitoring
→ **Updated Risk Score**
→ Premium adjustment / Coverage expansion or restriction

STEP3: Claims Adjustment

AI log + human approval record + cloud/process evidence
→ **Cause identification: AI vs Human vs Cloud**
→ Responsibility Ratio + Final Payout

SIMULATOR Link : <http://3.26.6.186/>

- **Step 1.** Choose an input mode
— Scenario, Manual, or Company Profile
- **Step 2.** Enter the values
 - Scenario — pick one of five incident cards (L1 / L1+L2 / L2 / L3 / L2+L3)
→ P values auto-set
 - Manual - adjust $P_{clintox} \cdot P_{medhal} \cdot \lambda$ to auto-compose P_{design} , then set $P_{process} \cdot P_{cloud}$ directly
 - Company Profile — enter revenue, R&D ratio, AI intensity, and cloud dependence
→ P values auto-calibrated per company
- **Step 3.** Read the result
— RiskScore, Grade (A-E), underwriting action, premium loading, deductible, and coverage limit are computed in real time

AEGIS-Rx / AI Cascading Risk Insurance [KO] [EN] [Simulate →]

Scenario Manual Company Profile

SCEN · L1
Design-stage error Hallucination + toxic candidate caught in R&D.

SCEN · L1+L2
Design → production Design defect combined with process miss.

SCEN · L2
Process AI miss Apparent normal, real defect → recall.

SCEN · L3
Cloud delay trigger No batch loss; parametric payout.

SCEN · L2+L3
Cloud + batch disposal Concurrent cloud outage + AI miss.

RISKSCORE
62.4 /100
D High Deduct. — Reinsurance recommended
P_design 0.400 P_process 0.816 P_cloud_adj 0.713

AEGIS-Rx / AI Cascading Risk Insurance [KO] [EN] [Simulate →]

Scenario **Manual** Company Profile

Adjust layer probabilities
P_design is auto-composed as $P_{clintox} \cdot (1 + \lambda \cdot P_{medhal})$. Default $\lambda = 0.5$.

P_clintox
L1a · Toxicity FN (threshold 0.3) 0.396

P_medhal
L1b · Hallucination FN (threshold 0.15) 0.020

λ
MedHal reliability penalty 0.50

COMPOSED P_DESIGN
 $= P_{clintox} \cdot (1 + \lambda \cdot P_{medhal})$ **0.400**

P_process
L2 · Missed Detection Rate 0.816

P_cloud
L3 · Hybrid Score · 0~1 0.800

RISKSCORE